

Track: HealthTech | Challenge ID: HT-03

Design an Intelligent Clinic Appointment and Queue Optimization System for Reducing Patient Waiting and Resource Congestion

Problem Description

Clinics often experience uneven appointment demand, walk-in arrivals, delays, and limited visibility into expected waiting time. The technical challenge is to design a queue-management prototype that combines appointments and walk-ins, allocates service slots, tracks queue state, and estimates waiting time using configurable service assumptions. The system should focus on operational scheduling rather than medical triage; any priority status used in the prototype must be assigned explicitly by authorized staff or simulated input rather than inferred from symptoms.

Technical Scope and Minimum Requirements

The platform should include appointment booking, provider or service selection, slot capacity, walk-in token generation, queue status, estimated wait time, and completion/cancellation updates. Teams should implement a scheduling or queue algorithm that responds to delays and no-shows. A staff dashboard and patient-facing status view are expected. Optional enhancements include overbooking limits, predicted service duration, or queue-load balancing across multiple service counters.

Expected Outcomes

A successful system will show how appointment and walk-in events affect the active queue and waiting-time estimates. Judges should be able to book slots, add walk-ins, mark a consultation as delayed or completed, and observe automatic queue recalculation. The platform should display current token, estimated wait, next appointments, provider load, and simple operational metrics such as average waiting time or number of delayed cases.

12-Hour Minimum Deliverable and Live Demonstration

Within 12 hours, teams should demonstrate one clinic with one or more service providers using simulated patient data. The live demo must include appointment creation, a walk-in, queue progression, a delay scenario, and updated wait-time estimates. Integration with hospital information systems and clinical decision support is outside the mandatory scope.